

running their own businesses and want to cater to their own set of clientele. The more the airlines, hotel chains and car rental companies are on the platform, the more they can create a comprehensive package for travellers, making travel more interconnected and seamless. That's what I mean when I emphasise the need for an open industry. Every player sees a benefit in being a participant in an open platform, which goes way beyond even open source.

Q. How and at what level is Amadeus using open source?

A. We leverage open source technologies quite significantly because again, in this entire complex and interconnected network, speed, agility, and innovation are the key essence to success for any of the players, including Amadeus, who provides the backbone. Now, if you look at the above said parameters, we are much better off using software that has been developed in an open community rather than trying to have our own proprietary system. That's why we are a big believer in open source and leverage several open source technologies on our platform.

And the good thing is that, now, a lot of these open source platforms are getting more sophisticated and wider in their reach. So, we benefit from that like any other player, and then we leverage these open source software in a responsible way to ensure security and privacy. That's how we play in the open source world.

Q. Where all are you using open source? Are there any broad packages or platforms that Amadeus is a big user of?

A. We have been running our own private cloud in our data centre ever since the company was created. And we are a pioneer in running a SaaS-based platform to scale because we

started doing this 37 years back, and we now run this entire platform in a virtualised setup.

So, we use Red Hat, OpenShift and OpenStack extensively. We are one of the premier partners for Red Hat, and we contribute a lot to the evolution of the Red Hat, OpenShift, Kubernetes, and OpenStack platforms. Then, we provide a lot of frameworks on the e-commerce side, which are wide-level products on which travel agencies and airline websites function. For that, we use many open source UI frameworks like Angular and React.

We even created our own proprietary framework, way back in 2009, and then made it open source. It was called ARIA templates (Amadeus Rich Internet Applications). Now, it's actively getting enhanced by the wider developer community. So we have open source frameworks.

We use a lot of Apache Kafka for streaming because it's a volume-intensive platform that we deal with. We use Hadoop Spark, etc, on the data platform that we have created, for gaining insights into the travel industry. So, I would say that open source goes across many spectrums of the travel industry as well as software engineering.

Q. How does your team evaluate which ones to bet upon? Do you carry out proof-of-concepts (PoCs) or some studies? Or is the decision just based on Gartner quadrants?

A. It's a mix of all of that. Of course, we constantly scout to see how the open source community is evolving. Now, more sophisticated open source platforms are coming in. We do our own PoCs once we identify and narrow some of these. And, of course, we participate in industry studies also.

But the key and fundamental aspect we look for is the platform's stability and readiness to scale. Being

a large platform with high availability and intensity of transactions, it's extremely important for our software to stand the test of volume, resilience and robustness.

So, we do experiment a lot and our global developer community is working to bring more and more open source into the Amadeus platform. We are extremely careful to make sure that it scales successfully because once we expose that and the entire industry—some 1000 airlines, 300,000 travel agencies, 100,000 hotel properties among others—starts using it, it better behave.

Q. Is there a centre of excellence or some team that looks after this whole domain of open source at Amadeus?

A. We have what we call a DevRel engineering team within the organisation. This is a transversal team that looks at making life better for our engineers and making them more productive. So when any of our engineers find some of these open source libraries or platforms that are available, they eventually funnel them into this DevRel community. It's made up of architects and principal engineers, and we operate it in an inner source kind of model. Once the software is identified, it goes to this community for evaluation, thus enabling the developer network to do PoCs and then come back to the community to report its findings.

The architecture community takes the final call to release it for wider use in two ways — one is just for our engineering community, and the second is for wider production use. The testing rigour is a lot more for the latter, because it's developed for general availability. We need our IP and legal teams to come in, and make sure that we are not violating any patents, etc. If it's for production use, then we bring other departments of the company into the assessment process.